

Doc link:-

https://docs.google.com/document/d/150mR_Q3ju3ujFz752uEocnvLjjQUg0IaNOMmmPwqLRU/edit?usp=sharing

chapter link:- <https://jax-ml.github.io/scaling-book/training/#what-do-we-mean-by-scaling>

Let's use LLaMA-2 13B as a basic model for this section. Here are the model details:

hyperparam	value
L	40
D	5,120
F	13824
N	40
K	40
H	128
V	32,000

Question 1: How many parameters does LLaMA-2 13B have (I know that's silly but do the math)? *Note that, as in [Transformer Math](#), LLaMA-3 has 3 big FFW matrices, two up-projection and one down-projection. We ignored the two "gating" einsum matrices in this section, but they behave the same as W_{in} in this section.*

My answer:-

$W_{in1}[D,F] = W_{in2}[D,F] = 7.1e7$

$W_{out}[F,D] = 7.1e7$

$W_{totaloutin1in2} = 2.2e8 * L = 9B$

$Vocab[V,D] = 2 * 1.64e8 = 0.32B$

$W_q[D,N,H] = W_k[D,K,H] = W_v[D,K,H] = 0.26e8$

$W_{kqvtotal} = 4 * 0.26e8 * L = 4B$

Question 2: Let's assume we're training with BS=16M tokens and using Adam. Ignoring parallelism for a moment, how much total memory is used by the model's parameters, optimizer state, and activations? Assume we store the parameters in bf16 and the optimizer state in fp32 and checkpoint activations three times per layer (after the three big matmuls).

My answer:-

in adam we have momentum and velocity matrix of same size as parameters

so parameters = 13B in bf16 = 26GB

optimizers is 13B*2 in fp32=104GB

for activation its $2 * \text{numlayer} * (\text{activation size of a mat}) = 2 * 40 * (BF + BF + BD)$ in bf16=42TB

Question 3: Assume we want to train with 32k sequence length and a total batch size of 3M tokens on a TPUv5p 16x16x16 slice. Assume we want to use bfloat16 weights and a float32 optimizer, as above.

1. Can we use pure data parallelism? Why or why not?
2. Can we use pure FSDP? Why or why not? With pure FSDP, how much memory will be used per device (assume we do gradient checkpointing only after the 3 big FFW matrices).
3. Can we use mixed FSDP + tensor parallelism? Why or why not? If so, what should X and Y be? How much memory will be stored per device? Using only roofline FLOPs estimates and ignoring attention, how long will each training step take at 40% MFU?

My answer:-

1 for pure batch $B/N > C/w_{\text{ici}} = 2550$ in this case $B/N = 732$ so not commune bound so we can use it. but model size is 130GB with optimizers so not possible

2 for FSDP the equation is same so no commune bound normally.

per device memory for just matrix replace 16M by 3M in Q2 = 7.85 TB/XYZ = 2GB

now for comm or compute if $B M_{xyz}/N > C/w_{\text{ici}} \Rightarrow B > 16 * 16 * 16 * 2550 / 3 = 3.4M$ token so commune bound

3

$B/N > \alpha^2 / M_x M_y F = 235$ so yes we can use it

$X_{\text{opt}} = \sqrt{B/F * M_x / M_y * N}$ $M_x = 2, M_y = 1$

memory is around 2 GB per device

total flops = 6* num parameters * model size, so time is totalflops/c/0.4=300ms

Xopt=1333

X=1024